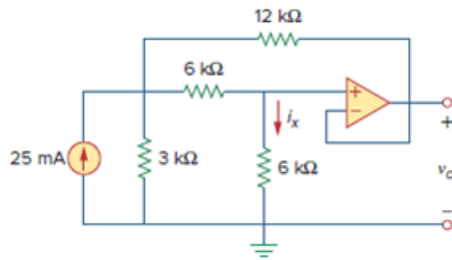


Problem

For the circuit in Fig. 5.69, find i_x .

Figure. 5.69:

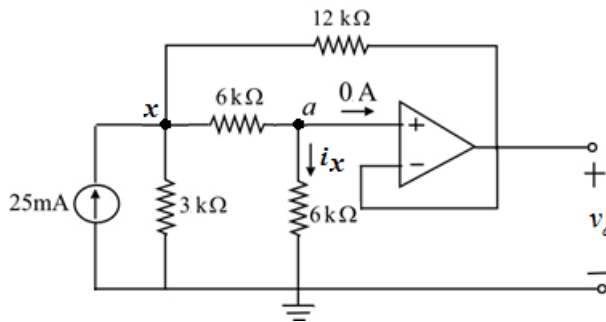


Step-by-step solution

Step 1 of 4

Refer to Figure 5.69 in the text book for the op-amp circuit.

Redraw the circuit by marking nodes.



Step 2 of 4

The op-amp circuit acts as a voltage follower.

$$v_a = v_o$$

Apply Kirchhoff's current law at node x.

$$25 \text{ mA} = \frac{v_x}{3 \text{ k}\Omega} + \frac{v_x - v_o}{6 \text{ k}\Omega} + \frac{v_x - v_o}{12 \text{ k}\Omega}$$

$$25 = \frac{v_x}{3} + \frac{v_x - v_o}{6} + \frac{v_x - v_o}{12}$$

$$25 = \frac{4v_x + 2v_x - 2v_o + v_x - v_o}{12}$$

$$7v_x - 3v_o = 300 \dots\dots (1)$$

Step 3 of 4

Apply Kirchhoff's current law at node a .

$$\frac{v_x - v_o}{6\text{ k}\Omega} = \frac{v_o}{6\text{ k}\Omega} + 0\text{ A}$$

$$v_x = 2v_o$$

Substitute $2v_o$ for v_x equation (1).

$$\begin{aligned} 300 &= 7(2v_o) - 3v_o \\ &= 11v_o \end{aligned}$$

$$v_o = \frac{300}{11}\text{ V}$$

Step 4 of 4

Apply ohm's law to find the current, i_x .

$$\begin{aligned} i_x &= \frac{v_o}{6\text{ k}\Omega} \\ &= \frac{\left(\frac{300}{11}\text{ V}\right)}{6\text{ k}\Omega} \\ &= 4.545\text{ mA} \end{aligned}$$

Thus, the current i_x in the $6\text{ k}\Omega$ resistor is 4.545 mA.